REFLOWECONOMY · TRACEABILITY MATRIX

ReflowEconomy traceability matrix

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/refloweconomy		
Source	docs/03b-traceability.md @ 9819cf9		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.2	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.1	2026-10-25	Amish Chadha	Generated by .kit/trace.py

ReflowEconomy traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 19 user requirements, 20 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001	PR-001	Feasibility matrix	RFE-CAL-001	Inspection (document review), TRL 3	Not met**: routes, ratings and reasons for 7 groups; hazard, export point and source columns still to add
UR-002	PR-002	Process recipes	RFE-CAL-001	Inspection against published processing data, TRL 3	Met on paper: four sourced recipes in `docs/playbook/recipes/`
UR-003	PR-003	Economics model	RFE-CAL-001	Analysis (RFE-CAL-001 section 10), TRL 3; review with an operating recycler at P5	Met on paper: result +0.17 USD per shift before rent; break-even product price 2.49 USD/kg; break-even input 100 kg per shift
UR-004	PR-004	Export rules	RFE-CAL-001	Inspection (document review), TRL 3	Not met**: principle and the cell packing rule only
UR-005	PR-005	Small site	RFE-CAL-001, RFE-BLD-001	Analysis of the model and RFE-DWG-001 (RFE-CAL-001 section 8), TRL 3; tape measure at TRL 4	Met on paper: 59.4 m2; aisle 1.57 m; headroom 2.02 m under the padded tray crossing; hot zone 1.00 m from stock (at the limit)
UR-006	PR-006	Throughput	RFE-CAL-001, RFE-STM-001	Analysis (RFE-CAL-001 section 3), TRL 3; measured rates at TRL 4	At risk**: shredder 3.79 h of 4 h at an assumed 15 kg/h (published 8.8 to 41.8 kg/h); press 2.06 sheets of capacity for 2; at the PR-016 intake threshold the shredder needs 4.26 h
UR-007	PR-007	Value kept local	RFE-CAL-001	Analysis (mass balance, RFE-CAL-001 section 2), TRL 3; site data at P5	Met on paper: 51.6 %
UR-008	PR-008	Disposal	RFE-CAL-001, RFE-STM-001	Analysis (mass balance, RFE-CAL-001 section 2), TRL 3; site data at P5	Not met** for the reference input: 28.4 %; 19.5 % for a load at the PR-016 threshold
UR-009	PR-009	Energy	RFE-CAL-001, RFE-BLD-001	Analysis (RFE-CAL-001 section 4), TRL 3; energy meter at TRL 4	Met on paper: 0.88 kWh/kg (45.6 kWh per shift; 0.83 before the STEM check fix, RFE-DEC-001 D2)
UR-016, UR-003	PR-010	Equipment cost	RFE-CAL-001, RFE-OPT-001	Analysis (priced BOM, RFE-CAL-001 section 9), TRL 3	Over the value-engineering target by USD 505: USD 25,505 indicative
UR-011	PR-011	Fume capture	RFE-CAL-001,	Analysis (RFE-CAL-001 section 6;	To verify at TRL 4: two enclosing hoods sized for 0.5 m/s (0.57 m3/s,

User requirement	Product requirement	Title	Evidence	Verification	Status
			RFE-BLD-001	RFE-STM-001 S10), TRL 3; anemometer and air monitoring at TRL 4	fan 1.00 kW of 1.5 kW); exposure needs air monitoring
UR-012	PR-012	Machine, heat, noise and electrical safety	RFE-CAL-001, RFE-BLD-001	Analysis (RFE-CAL-001 section 7; RFE-STM-001 S15 to S17), TRL 3; measurement at TRL 4	At risk**: reach 1,060 mm through a 460 x 120 mm slot (passes; slot lowered from 160 mm, RFE-DEC-001 D3); LEX 79 dB(A) enclosed on an assumed sound power; the EN 12012 type C standard for size reduction machines still to check
UR-013	PR-013	Passport on every lot	RFE-CAL-001, RFE-BLD-001	Test (schema validation of the example records, RFE-CAL-001 section 11), TRL 3	Met on paper: 3 of 3 examples valid
UR-013	PR-014	Trace to intake	RFE-CAL-001	Inspection of the schema, TRL 3	Met by design: v0.2 requires both
UR-014	PR-015	Quick to fill in	RFE-CAL-001, RFE-BLD-001	Test (timed trial with operators), TRL 4	To verify at TRL 4: needs a form and a timed trial
UR-008, UR-019	PR-016	Intake quality rule	RFE-CAL-001	Analysis (RFE-CAL-001 section 2), TRL 3; intake records at P5	Met on paper: threshold 10 % (limit 10.6 %); a cleaner load carries more plastic and pushes PR-006 past its 4 h block
UR-010	PR-017	Supply current	RFE-CAL-001, RFE-BLD-001, RFE-STM-001	Analysis (RFE-CAL-001 section 5; RFE-STM-001 S14), TRL 3; clamp meter at TRL 4	Not met**: block A about 55 A running (72 A at full load); block B 36 A. Supply decision proposed, awaiting Amish (RFE-DEC-001 D1)
UR-015	PR-018	Local fabrication	RFE-CAL-001, RFE-STM-001	Inspection of the model, drawings and build plan (RFE-BLD-001), TRL 3	Met by design: 11 made components with making sketches; detailed model at D3 in standard stock (RFE-CAL-001 section 8)
UR-018	PR-019	Fire and lithium cells	RFE-CAL-001	Inspection of the model, BOM lines 10 and 12 and the build plan stop S5, TRL 3	Met by design (RFE-STM-001 S27; RFE-CAL-001 section 8)
UR-017	PR-020	Registration	RFE-CAL-001,	Inspection (RFE-REG-001; RFE-CAL-001 section 11),	To verify at TRL 4: listed for Maharashtra (MPCB registration on

User requirement	Product requirement	Title	Evidence	Verification	Status
			RFE-STM-001	TRL 3; registration itself at P5	the CPCB portal, under 200 t a year band; Air and Water Act consents)

Gaps

None.